

Supplementary material: A finite weighted-graph consistency check

Chatchawan Panraksa

September 10, 2026

This supplement accompanies *Effective boundary conditions and spectral asymptotics for shrinking exterior Dirichlet layers*. It records an independent finite-dimensional check of exterior minimization and the bounded-perturbation estimates in Appendix A of that manuscript. The graph is defined directly, with counting measure on its vertices; its eigenvalues are not approximations to the continuum eigenvalues studied in the paper.

S.1 The graph and exterior minimization

We work in $\mathbb{R}^8$ with the Euclidean inner product. The interior vertices are $x_i = 0.08 + 0.12(i-1)$, $1 \leq i \leq 8$, and the exterior vertices and weights are

$$(z_1, z_2, z_3, z_4) = (-1.4, -0.6, 1.5, 2.2), \quad (\omega_1, \omega_2, \omega_3, \omega_4) = (0.02, 0.03, 0.025, 0.02).$$

Set $s = 0.65$, $c = 0.7$, $K_{ij} = |x_i - z_j|^{-1-2s}$, $k_j = (K_{ij})_{i=1}^8$, and $m_j = \sum_{i=1}^8 K_{ij}$. The interior graph has weight $1.3 + 0.1(i-1)$ on the edge joining i and $i+1$, together with weight $0.015|x_i - x_\ell|^{-1-2s}$ on every pair $i \neq \ell$. The weights add on neighbouring pairs. If W is this symmetric interior weight matrix with zero diagonal, let $L = \text{diag}(W\mathbf{1}) - W$.

For an interior vector $u \in \mathbb{R}^8$ and exterior values $v \in \mathbb{R}^4$, the energy is

$$E(u, v) = u^\top L u + c \sum_{j=1}^4 \omega_j \sum_{i=1}^8 K_{ij} (u_i - v_j)^2.$$

Minimization over each v_j gives $v_j = k_j^\top u / m_j$. Hence the reduced energy is $u^\top A_N u$, where

$$A_N = L + c \sum_{j=1}^4 \omega_j \left(\text{diag}(k_j) - \frac{k_j k_j^\top}{m_j} \right). \quad (\text{S.1})$$

This matrix is also the Schur complement of the exterior block in the full energy matrix. Its normalized nullvector is $e = \mathbf{1}/\sqrt{8}$. The computed positive spectral gap is $g \simeq 1.29156104514$.

S.2 Spectral checks and reproduction

We impose a penalty on the exterior vertices z_2 and z_3 by setting

$$B = c \sum_{j=2}^3 \omega_j \frac{k_j k_j^\top}{m_j}, \quad A_\tau = A_N + \tau B, \quad \eta_\tau = \tau c \max_{1 \leq i \leq 8} \sum_{j=2}^3 \omega_j K_{ij}. \quad (\text{S.2})$$

The parameter τ scales a fixed matrix penalty; it is independent of the collar width ε in the manuscript. For $0 \leq \tau \leq 1$, it also has an exact graph interpretation: split each selected

exterior vertex into a free part of weight $1 - \tau$ and a part of weight τ held at zero. Eliminating the free parts leaves the matrix A_τ . The script checks this identity while retaining the full original exterior edge mass in the interior diagonal.

Let $Q = I - ee^\top$ and let A_Q denote the restriction of A_N to $e^\perp$. For $\lambda_1(\tau) = \lambda_{\min}(A_\tau)$, define

$$a_\tau = e^\top(\tau B)e, \quad r_\tau = Q(\tau B)e.$$

The finite-dimensional specialization of Proposition A.2 in the accompanying manuscript gives

$$0 \leq a_\tau - \lambda_1(\tau) \leq \frac{\|r_\tau\|^2}{g - \eta_\tau}$$

and the second-order remainder estimate

$$\begin{aligned} \mathcal{R}_\tau &= \lambda_1(\tau) - a_\tau + r_\tau^\top A_Q^{-1} r_\tau, \\ |\mathcal{R}_\tau| &\leq \frac{\eta_\tau \|r_\tau\|^2}{g(g - \eta_\tau)}. \end{aligned} \tag{S.3}$$

Here $\eta_1 \simeq 0.06927310317 < g/2$, so these bounds apply at every tested value $\tau = 2^{-k}$, $0 \leq k \leq 5$. Table S.1 reports the first- and second-order errors at these six values.

τ	$a_\tau - \lambda_1(\tau)$	$ \mathcal{R}_\tau $	Bound for $ \mathcal{R}_\tau $
1	4.075×10^{-5}	5.252×10^{-7}	5.957×10^{-6}
0.5	1.012×10^{-5}	6.523×10^{-8}	7.241×10^{-7}
0.25	2.522×10^{-6}	8.128×10^{-9}	8.928×10^{-8}
0.125	6.296×10^{-7}	1.014×10^{-9}	1.108×10^{-8}
0.0625	1.573×10^{-7}	1.267×10^{-10}	1.381×10^{-9}
0.03125	3.930×10^{-8}	1.583×10^{-11}	1.723×10^{-10}

Table S.1: Errors for the finite weighted-graph comparison, with perturbation τB . The second-order remainder and its bound are defined in (S.3). All bounds use the graph’s own Neumann gap and defect parameter.

The computations used Python 3.12.14 and NumPy 2.3.5. The single random vector used for the energy identity has fixed seed 1729; all graph weights and perturbations are deterministic.

The verification script was developed with OpenAI Codex assistance. It also checks the bound $\|\tau B\| \leq \eta_\tau$, all eigenvalue-shift inequalities, the shifted-resolvent estimate at shift 1, the semigroup estimate at time 0.8, and the form-based ground-state bound. These matrix comparisons use absolute tolerance 3×10^{-11} . The recorded Schur-complement discrepancy, minimized-energy discrepancy, and constant-nullvector residual are approximately 6.94×10^{-18} , 0, and 1.20×10^{-15} , respectively.

The `verification` directory in the supplementary code archive contains the script `verify.py`, its recorded output `results.json`, and the table generator `make_table.py`. Running `python3 build.py` from the extracted code archive reruns the graph checks and the manuscript’s interval calculations and regenerates the numerical tables and plot data. The graph experiment checks the finite implementation and illustrates the bounded-perturbation estimates; it does not establish a continuum discretization error or certify continuum eigenvalues.